

34870 Lab D - Loudspeaker impedance and near field

Room 026 – Building 354

Purpose:

- 1) Measuring the impedances of the driver units in the loudspeaker system to be used in the project.
- 2) Deriving the Thiele-Small parameters (f_s , Q_{ms} , Q_{ts} , Q_{es} and V_{as}) for the woofer unit.
- 3) Illustrating low frequency effects of a boxed loudspeaker.

Warnings:

- Loudspeaker units are fragile, especially the tweeter, which can be damaged by even a light touch. Please be careful in not touching the drivers when handling the loudspeaker boxes.
- Your loudspeaker system consists of two boxes: a woofer box and a mid/tweeter box. There are six systems named with letters A to F. The mid/tweeter box has a vent that must be kept closed; only the vents in the woofer box are used for low frequency design. You must use the same system for the rest of the course. Some groups may have to share systems: avoid booking a slot in parallel with another group using your system.
- Especially for the near-field measurements in section 3, the influence of room modes can be important. Absorbing material is distributed over the walls of the lab to mitigate it, but modes might still be visible in the near-field responses. You can measure the room dimensions and calculate modes to find their frequencies, as the room is rectangular.

Equipment:

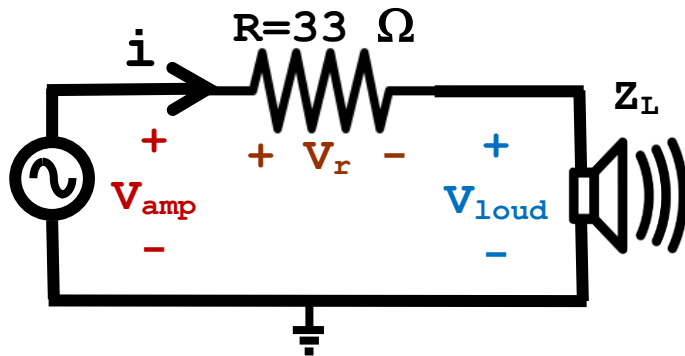
- Desktop PC,
- Multimeter,
- Two vent plugs,
- National Instruments NI-4431 data acquisition card
- Circuit connection board + resistors
- Nexus microphone amplifier
- Bruel & Kjaer microphone or GRAS microphone (free-field or pressure-field)
- 3-way loudspeaker system corresponding to your group (see booking datasheet),
- Audio amplifier.

Part 1 - DC resistance:

Measure the DC resistance R_E of each driver with the multimeter.

Part 2 – Electrical input impedance

The input electrical impedance of the loudspeaker is measured as follows. The amplifier output is connected to the loudspeaker through a circuit connection board that allows placing a resistor in series. The board has also connectors for measuring the amplifier (V_{amp}) and loudspeaker voltages (V_{loud}), as shown in the sketch.



The output channel A00 of the NI 4431 is connected to one of the inputs of the audio amplifier, which is then selected in its front panel.

The amplifier output (V_{amp}) is fed into channel AI0 of the NI 4431, and the loudspeaker excitation signal (V_{loud}) is fed into channel AI1 of the NI 4431.

A multitoned signal will be generated to cover the requested frequency range. The NI 4431 card can accept up to +/- 10 volt on each input channel. The loudspeaker impedance Z_L is calculated from the amplifier output voltage V_{amp} and the loudspeaker voltage V_{loud} , using the relation (deduce it yourself by analysing the circuit):

$$Z_L = R \frac{\frac{V_{loud}}{V_{amp}}}{1 - \frac{V_{loud}}{V_{amp}}}$$

Note that the voltage V_r cannot be measured directly on the resistor because the NI 4431 card inputs are grounded. The function `meas_mag_spec2_LabD` is used for measuring:

```
[fn,specn,f,spec,ch]=meas_mag_spec2_LabD(f1,f2,n_oct,fres,Nav)
```

Input parameters:

f1: lower measurement frequency
 f2: upper measurement frequency
 n_oct: number of measurement frequencies per octave
 fres: frequency resolution of spectrum, signal period $T=1/fres$
 Nav: Number of periods to average over

The outputs are:

fn: signal frequency vector
 specn: spectrum values for fn
 f: full frequency vector
 spec: full resolution spectrum
 ch: average time signals

`specn` contains the spectrum at the signal frequencies only. `spec` contains the full spectrum and can reveal the background noise level and any distortion in the measurement.

As in previous exercises, the frequency response V_{loud}/V_{amp} can be calculated by dividing the two columns of `specn` elementwise: `specn(:,2)./specn(:,1)`, or the inverse if the cables are switched. As a check, the input impedance Z_L should tend to the DC resistance R_E at low frequencies.

2a. Electrical impedance measurement for the crossover project

In this section the input electrical impedances of the loudspeaker project units are measured. Connect a 33 ohm resistor in the circuit connection board. Write a Matlab script that calls *meas_mag_spec2_LabD()* with the parameters given below. The script must calculate the impedance using the expression in the previous section. The plots produced by the measurement function are just for checking; you must produce your own figures.

Adjust the gain of the amplifier to get a clean and smooth measurement. Note that the sound level from the loudspeaker is expected to be low due to the resistor: what matters is the quality of the input impedance result. If the signal level is too low, the measurement is corrupted by noise, and if it is too high it will be corrupted by non-linear distortion from the loudspeaker. See examples of such measurements in Appendix A. Usually, a rms value of the loudspeaker voltage in the 0.1 volt range is a good starting point. *meas_mag_spec2_LabD* calculates the rms value of the two input channels.

2a1) Tweeter:

Use the following measurement parameters:

```
f1=100
f2=24000
n_oct=48
fres=0.125
Nav=1
```

Calculate, plot and save the complex impedance.

2a2) Midrange:

Use (the rest remain the same):

```
f1=20
f2=20000
```

Calculate, plot and save the complex impedance.

2a3) Woofer: Open vent.

Use (the rest remain the same):

```
f1=1
f2=10000
```

This measurement can be done along with section 2b below.

2b. Thiele-Small parameters of the woofer

The T-S parameters of the woofer driver can be obtained from the following measurement. They can be used for the low-frequency design of the woofer box.

Optional reproducibility check: you will get several 33 ohm resistors with a tolerance of 5% (code J). The variations of the resistance value could lead to noticeably different results. You may experiment by changing the resistor and repeating the T-S parameter calculation. It is important to be aware of this issue as you will also use components with tolerances for your passive crossovers in the project.

2b1) Free air

- Remove the back cover of the woofer box. Take out any filling material.
- Measure the woofer input electrical impedance. Do not place the unit facing a reflective close surface. Store the results in the computer.
- Identify the resonance frequency f_s and Q -values, as shown in Appendix B.

2b2) Closed box

- Close the box and block the vent. Keep the box free of filling material for a more precise determination of the box volume. Replace the filling material after your measurements as it is needed to damp box resonant modes at higher frequencies.
- Measure the input impedance. Do not place the unit facing a reflective close surface. Store the results in the computer.
- Identify the resonance frequency, f_c and corresponding Q -values using the method described in Appendix B and course slides.

2b3) Box volume

- Measure the inner dimensions of the box and use them to calculate V_{ab} (effective volume of the box). V_{ab} will be used for obtaining V_{as} . If there is no filling (recommended in his case), V_{ab} is equal to the physical volume of the box. No need to consider the volumes of vents and unit inside the box.

2b4) Vent influence

- Uncover the vent(s) and repeat the measurement for several tube lengths. Take note of the lengths.
- Find the vent resonance frequencies, f_b , from the input impedances, and through calculation using box volume, tube lengths and tube radii. Note that your box has two physical vents that can be combined into one (use theory). You may block one of them.

Part 3 - Near Field responses of the woofer unit and the vent

Since anechoic conditions are hard to fulfil below 100 Hz in anechoic chambers (such as the one in the labs), the near field technique is often used at low frequencies. At low frequencies the radiation impedance is dominated by the mass of air, M_{A1} , and the sound pressure in front of the diaphragm can therefore be assumed to be:

$$p_{near} = j\omega M_{A1} U, \quad M_{A1} = \frac{8\rho}{3\pi^2 a} \quad (\text{Assuming baffled conditions})$$

The near field pressure, therefore, has the same frequency response as the far field pressure, but the level is of course higher. Note that, since the vent radius rarely is the same as the woofer radius, the levels of the *two near field pressures are not directly comparable* and can only be added to give the far field pressure after appropriate scaling.

Measurement procedure:

- Connect the microphone to the Nexus amplifier.
- Make sure the microphone is connected before turning on the amplifier. This forces the Nexus amplifier to autodetect the microphone sensitivity.
- Set the output gain to 1 V/Pa.
- Connect the output from the Nexus to input channel 1 on the NI 4431.
- Replace the resistor in the connection board by a short circuit (join the two terminals).

Use the same measurement function and input parameters as in the case of the woofer input impedance measurement. The function produces Matlab figures showing signals, for control purposes. Calculate the transfer function between the microphone output and the loudspeaker signal: $H_w = \text{specn}(:,2)/\text{specn}(:,1)$.

You will need to run some trial measurements to find the proper volume levels. Adjust the volume control in the audio amplifier (starting very low) until the amplitude of the measured time signals are not much smaller than one, but never over one. Use the plots generated by the measurement function for verification. Since the background noise can be a problem at lower frequencies you may try to reduce it by increasing the `Nav` input parameter to 4 or more.

3a) Vent influence

Measure the near field sound pressure response at the diaphragm and at the vent for at least three different tube lengths. Find f_b from the response and compare with the results from 2b4 and your analytical calculations.



Use of the results of Lab D in the loudspeaker project

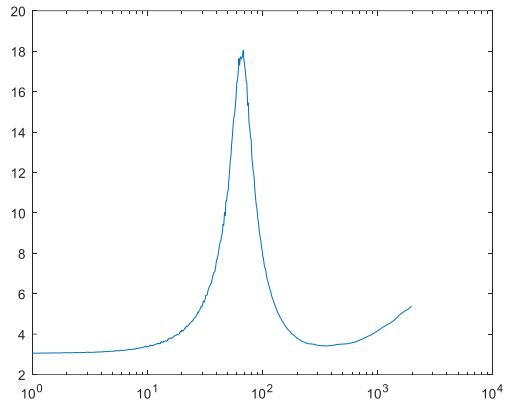
The results from this exercise will be used in the loudspeaker project, where you will need:

1. Thiele-Small parameters of the woofer unit for the low frequency design of woofer box with vent. You are expected to produce a circuit analogy in the project.
2. The near-field measurements will be used to validate the circuit analogy representing the woofer box. You may need to repeat this as a part of the low-frequency design of the project woofer box. You learn the technique in Lab D. Both Lab D and Lab E setups remain accessible during the project.
3. Impedance responses of midrange and tweeter units for the calculation of the box ensemble in connection with cross-over filters design. The woofer input impedance is not needed for this purpose as its cross-over will be digital.

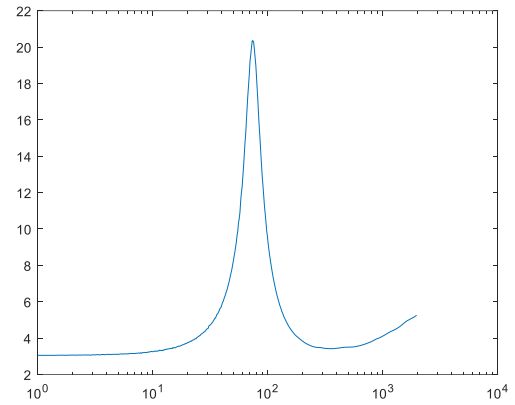
Evaluation of the exercise

The exercise work is made in groups of ideally three students. However, the evaluation is individual using a quiz in DTU Learn. The responses do not need to be the same for all students in the group. The quiz covers Lab D and Lab E. Submit the quiz as soon as possible and **no later than the deadline Monday 26th October**.

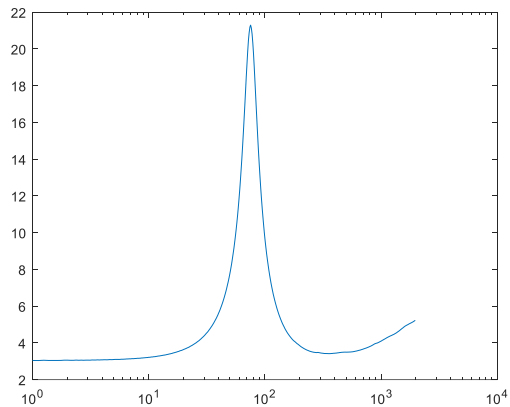
Appendix A: Effect of the input signal level on the impedance response:



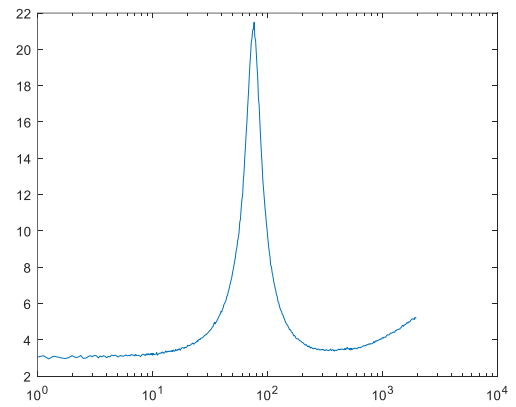
Measurement at 0.7 volt rms at loudspeaker



Measurement at 0.06 volt rms at loudspeaker



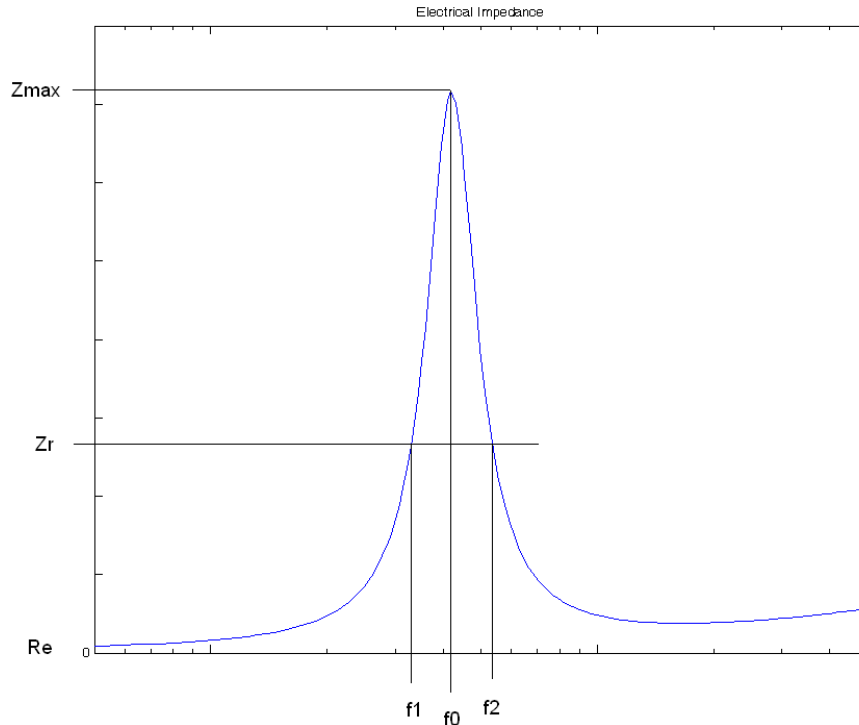
Measurement at 0.01 volt rms at loudspeaker



Measurement at 0.005 volt rms at loudspeaker

Appendix B:

Identifying the resonance frequency and Q factor from the impedance curve:



The electrical impedance of the loudspeaker is

$$Z_{ET} = R_E + \frac{(Bl)^2}{j\omega M_{MS} + R_{MS} + 1/(j\omega C_{MS})}, \quad Z_{MAX} = R_E + \frac{(Bl)^2}{R_{MS}}$$

The figure above shows the magnitude of Z_{ET} . From the ratio between the maximum impedance at resonance and the DC resistance of the voice coil

$$r_c = \frac{Z_{MAX}}{R_E} = 1 + \frac{(Bl)^2}{R_E R_{MS}}$$

the Q factors can be found by identifying the two frequencies for which:

$$Z_r = |Z_E(f_1)| = |Z_E(f_2)| = \sqrt{R_E Z_{MAX}}$$

$$Q_M = \frac{f_0 \sqrt{r_c}}{f_2 - f_1}$$

$$Q_E = \frac{Q_M}{r_c - 1}$$

$$Q_T = \frac{Q_M}{r_c}$$

Note that the method works for the loudspeaker in free air, baffled or in a closed box. It will not work with the input impedance of a vented box. Also notice that the equivalent volume V_{as} needs to be deduced from two measurements of the resonance frequency for baffled and closed box conditions. The procedure has been explained in Lecture 8.